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Purge stand

Abstract

A purging device includes a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body. A base is positioned adjacent the first end of the main body, a testing port is disposed along the main body of the purging device, and a control valve is disposed between the inlet of the main body and the outlet of the main body.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/215,610, filed on Jun. 28, 2021. The entire disclosure of the above application is incorporated herein by reference.

FIELD
(1) The present invention relates generally to gas purging systems and, more particularly, to a device, kit, and method for purging natural gas.
INTRODUCTION
(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) It is common to purge gas from piping, appliances, or other confined spaces. As one example, air from a newly constructed natural gas pipeline must be purged prior to filling the pipeline with natural gas in order to achieve one-hundred percent gas at all delivery points. Purging air from the natural gas pipeline militates against pilot lights and appliances being extinguished of their flames and the formation of a hazardous combination of air and natural gas that may be explosive or flammable. Additionally, gas purging may be necessary when repairing sections of existing pipeline. Proper above ground height as well as safe distances from buildings and overhangs is necessary when purging gas.
(4) Known devices used for purging pipelines are often unable to safely expel gas from a pipeline and are located too close to a residential building or overhanging structure creating dangerous situations due to proximity. In addition, test ports used to measure the level of combustible gas within the gas system may create undesirable Venturi effects. Known purging devices are also not adaptable in various settings and on variable terrain.
(5) In one particular example, U.S. Pat. No. 6,056,004 teaches a portable compression system for pipeline purging. However, the device does not safely and effectively expel dangerous gas away from the pipeline at a safe distance and is not adaptable or customizable in different surroundings. In another example, U.S. patent application Ser. No. 13/151,364 teaches a gas purging device and method including a gas purge burner used to burn off gas leaving a pipeline. However, the device does not address various safety concerns relating to purging a gas line and is cumbersome to use. In yet another example, U.S. patent application Ser. No. 17/480,338 teaches a gas purging apparatus that expels gas without discharge. However, the apparatus is difficult to use and not adaptable in various settings.
(6) Accordingly, there is a continuing need for a purging device that is safe, durable, modular, and adjustable. Desirably, the purging device would be easy and efficient to use and transport, inexpensive to manufacture, and configured to work in a variety of settings.

SUMMARY

(7) In concordance with the instant disclosure, a purging device that is safe, durable, modular, adjustable, easy and efficient to use and transport, inexpensive to manufacture, and configured to work in a variety of settings, has surprisingly been discovered.
(8) In certain embodiments, a purging device includes a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body. A base is positioned adjacent the first end of the main body, a testing port is disposed along the main body of the purging device, and a control valve is disposed between the inlet of the main body and the outlet of the main body.
(9) In another embodiment, a gas purging kit includes a purging device including a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body. The purging device includes a base positioned adjacent the first end of the main body, a testing port disposed along the main body of the purging device, and a control valve disposed between the inlet of the main body and the outlet of the main body. The gas purging kit further includes a flexible hose adapted to connect to the inlet of the main body, an extender adapted to connect to the outlet of the main body, at least one safety cone, and a carrying case.
(10) In another embodiment, a method of using a gas purging kit comprises the step of providing a gas purging kit. The gas purging kit includes a purging device including a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body. A base is positioned adjacent the first end of the main body, a testing port is disposed along the main body of the purging device, and a control valve is disposed between the inlet of the main body and the outlet of the main body. The gas purging kit further includes a flexible hose adapted to connect to the inlet of the main body, an extender adapted to connect to the outlet of the main body, at least one safety cone, and a carrying case. The method further comprises the steps of identifying a gas system for purging, assembling the purging device, connecting the inlet of the

purging device to the gas system, and purging a gas from the gas system.
(11) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS
(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** is a top perspective view of a purging device, according to an embodiment of the present disclosure;
(3) FIG. **2** is a top perspective view of the purging device of FIG. **1** shown in use;
(4) FIG. **3** is a top perspective view of a testing port and a control valve of the purging device of FIG. **1**;
(5) FIG. **4** is a top perspective view of a gas purging kit, according to an embodiment of the present disclosure; and
(6) FIG. **5** is a method of using a gas purging kit, according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
(7) The following description of technology is merely exemplary in nature of the subject matter, manufacture and use of one or more inventions, and is not intended to limit the scope, application, or uses of any specific invention claimed in this application or in such other applications as may be filed claiming priority to this application, or patents issuing therefrom. Regarding methods disclosed, the order of the steps presented is exemplary in nature, and thus, the order of the steps can be different in various embodiments, including where certain steps can be simultaneously performed, unless expressly stated otherwise. “A” and “an” as used herein indicate “at least one” of the item is present; a plurality of such items may be present, when possible. Except where otherwise expressly indicated, all numerical quantities in this description are to be understood as modified by the word “about” and all geometric and spatial descriptors are to be understood as modified by the word “substantially” in describing the broadest scope of the technology. “About” when applied to numerical values indicates that the calculation or the measurement allows some slight imprecision in the value (with some approach to exactness in the value; approximately or reasonably close to the value; nearly). If, for some reason, the imprecision provided by “about” and/or “substantially” is not otherwise understood in the art with this ordinary meaning, then “about” and/or “substantially” as used herein indicates at least variations that may arise from ordinary methods of measuring or using such parameters.
(8) Although the open-ended term “comprising,” as a synonym of non-restrictive terms such as including, containing, or having, is used herein to describe and claim embodiments of the present technology, embodiments may alternatively be described using more limiting terms such as “consisting of” or “consisting essentially of.” Thus, for any given embodiment reciting materials, components, or process steps, the present technology also specifically includes embodiments consisting of, or consisting essentially of, such materials, components, or process steps excluding additional materials, components or processes (for consisting of) and excluding additional materials, components or processes affecting the significant properties of the embodiment (for consisting essentially of), even though such additional materials, components or processes are not explicitly recited in this application. For example, recitation of a composition or process reciting elements A, B and C specifically envisions embodiments consisting of, and consisting essentially of, A, B and C, excluding an element D that may be recited in the art, even though element D is not

explicitly described as being excluded herein.

(9) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(10) Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

(11) Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(12) With reference to FIGS. **1-3**, a purging device **100** is shown. The purging device **100** may be configured to connect to a gas system **102** allowing a technician to purge gas **104** from the gas system **102**. As one non-limiting example, the gas system **102** may be a natural gas pipeline. The purging device **100**, according to certain embodiments, may include a main body **106**, a base **108**, a testing port **110**, and a control valve **112**.

(13) The main body **106** may be a conduit or other enclosed channel configured to receive gas **104** through an inlet **114** disposed at a first end **116** of the main body **106** and expel gas through an outlet **118** disposed at a second end **120** of the main body **106**. The main body **106** may have any desirable shape, length, diameter, and thickness, as determined by one of skill in the art. As non-limiting examples, in certain embodiments, the main body **106** may have a diameter of three-quarters of an inch, one inch, or two inches.

(14) Any material or combination of materials such as metal, plastic, composite, and wood may be used to fabricate the main body **106** of the purging device **100**. It should be appreciated that a skilled artisan may employ materials and dimensions that match particular industrial standards and regulations, as needed. The main body **106** may have a height **122** that is fixed or adjustable. In certain more particular embodiments, the main body **106** may be fabricated using steel and capable of withstanding line pressure up to 125 PSI. In certain more particular embodiments, the main body **106** may be made using schedule 40 piping. Advantageously, the purging device **100** is lightweight, easy to transport, and inexpensive to manufacture. In another more particular embodiment, the main body **106** may be fabricated using steel and capable of withstanding line pressure greater than 125 PSI.

(15) The inlet **114** may be configured to receive gas **104**, such as air or natural gas, from the gas system **102**. The inlet **114** may be integral with the main body **106** of the purging device **100** or

may be permanently or removably connected to the main body **106**, as determined by one of skill in the art. According to certain more particular embodiments, the inlet **114** may be disposed adjacent to a ground surface **124** to facilitate an easy, safe connection with the gas system **102**. However, a skilled artisan may position the inlet **114** anywhere on or along the main body **106** of the purging device **100** between the first end **116** and the outlet **118**.

(16) According to certain embodiments, the inlet **114** may be an opening defined by the first end **116** of the main body **106** (not shown). Alternatively, in certain more particular embodiments, the inlet **114** may extend outwardly from a sidewall **126** of the main body **106**, as shown in FIGS. **1-4**. As one non-limiting example, the inlet **114** may be a conduit extending outwardly at a 90-degree angle from the sidewall **126** of the main body **106** adjacent the first end **116** of the main body **106**. Any material or combination of materials such as metal, plastic, composite, and wood may be used to fabricate the inlet **114**. In certain more particular embodiments, steel may be used to form the inlet **114** or a portion of the inlet **114**.

(17) In certain embodiments, as shown in FIG. **2**, the purging device **100** may include a flexible hose **128** adapted to connect to the inlet **114** and be in fluid communication with the purging device **100** and the gas system **102**. The flexible hose **128** may be made from any durable material capable of withstanding pressure from the gas system **102**. As non-limiting examples, nylon, polyurethane, polyethylene, PVC, and synthetic or natural rubber may be used. A skilled artisan may select any suitable length and diameter for the flexible hose **128**. In one more particular embodiment, the flexible hose **128** may be 12 feet in length. It should be appreciated that a skilled artisan may achieve fluid communication between the purging device **100** and the gas system **102** by employing any number of components and methods, within the scope of this disclosure.

(18) The flexible hose **128** may be integral with or permanently or removably connected to the inlet **114** at a first end **130** of the flexible hose **128** using any suitable purge connecting means **132**. As non-limiting examples, the flexible hose **128** may be removably connected to the inlet **114** using a threaded connection, a snap fit, a friction fit, or a quick connect fitting. The flexible hose **128** may be configured to removably connect to the gas system **102** at a second end **134** of the flexible hose **128** using one or more gas connecting means **136**. In certain embodiments, the flexible hose **128** may include a variety of interchangeable gas connecting means **136** for connecting the second end **134** of the flexible hose **128** to a variety of gas systems **102**, as needed. Advantageously, the flexible hose **128** allows the technician to purge gas **104** at a safe distance from the gas system **102**. Additionally, the flexible hose **128** is lightweight and flexible for easy storage and transport and adaptable in variable settings and on variable terrain.

(19) The outlet **118** may be configured to expel gas **104** into an external environment. In one non-limiting example, the outlet **118** may direct air from the gas system **102** to achieve a higher natural gas level in the gas system **102**. According to certain more particular embodiments, the outlet **118** may be positioned to expel gas upwardly away from the ground surface **124**. However, a skilled artisan may position the outlet **118** anywhere on the main body **106** between the inlet **114** and the second end **120** of the main body **106**.

(20) In certain embodiments, the outlet **118** may be integral with the main body **106** of the purging device **100** or may be permanently or removably connected to the main body **106**, as determined by one of skill in the art. According to certain more particular embodiments, the outlet **118** may be defined by an opening **138** disposed at the second end **120** of the main body **106**. Alternatively, in certain embodiments, the outlet **118** may extend outwardly from the sidewall **126** of the main body **106** (not shown). Any material or combination of materials such as metal, plastic, composite, steel, and wood may be used to form or define the outlet **118**.

(21) With renewed reference to FIG. **2**, the outlet **118** may be configured to receive at least one extender **140** according to certain embodiments. The extender **140** may be configured to increase the height **122** of the main body **106** to militate against gas **104** escaping from the outlet **118** and contacting the technician or creating other hazardous conditions. The extender **140**, in one non-

limiting example, may be a conduit or other enclosed channel configured to receive the gas **104** from the second end **120** of the main body **106** at a first end **142** of the extender **140** and expel the gas **104** from a second end **144** of the extender **140** positioned at a distance from the purging device **100** and the gas system **102**. The extender **140** may be removably connected to the outlet **118** of the main body **106** using a threaded connection, a snap fit, a friction fit, or a quick connect fitting, as non-limiting examples.

(22) The extender **140** may have any desired shape, length, diameter, and thickness, as determined by one of skill in the art. The extender **140** may have a height **146** that is fixed or adjustable. Any material or combination of materials such as metal, plastic, composite, and wood may be used to fabricate the extender **140**. It should be appreciated that a skilled artisan may employ materials and dimensions that match particular industrial standards and regulations, as needed. In certain more particular embodiments, the extender **140** may be fabricated using aluminum. In certain more particular embodiments, the height **122** of the main body **106** combined with the height **146** of the extender **140** may equal one of seven feet and ten feet, as non-limiting examples. Advantageously, the extender **140** allows the technician to safely purge gas **104** at a customizable height above ground to prevent dangerous conditions, protect the technician and any other workers nearby, and meet applicable industry and regulatory standards.

(23) The main body **106** of the purging device **100** may include additional features such as a securing element **148**, as one non-limiting example. The securing element **148** may be integral with the main body **106** or permanently or removably connected to the main body **106**. The securing element **148** may be configured to connect to a strap or other tie-down means (not shown). The strap or other tie-down means may be configured to be connected to the ground surface **124** and used to secure and stabilize the purging device **100** with respect to the ground surface **124**. In one more particular example, a plurality of securing elements **148** may extend outwardly from the sidewall **126** of the main body **106**. Each securing element **148** may be adapted to receive a strap, rope, or other tie-down means adapted to connect to the securing elements **148** and to the ground surface **124**. Any suitable material and configuration may be used to form the securing element **148**. It should be appreciated that a skilled artisan may further stabilize the purging device **100** relative to the ground surface **124** by employing any number of components and methods, within the scope of this disclosure.

(24) A grounding rod and cable **150** may be permanently or removably connected to the main body **106** of the purging device **100** and configured to provide a low resistance electrical path to the ground surface **124**, according to certain embodiments. A skilled artisan may use any suitable grounding rod and cable **150**, as desired. The main body **106** may include one or more attachment means **152** for securing and storing the grounding rod and cable **150** with the purging device **100** when the grounding rod and cable **150** are not in use. Advantageously, the grounding rod and cable **150** may reroute unwanted static electricity to the ground surface **124** to militate against gas **104** being ignited from the static electricity.

(25) The base **108** may be integral with the first end **116** of the main body **106** of the purging device **100** or may be permanently or removably connected to the first end **116** of the main body **106** and may be configured to stabilize the purging device **100** during use. Any suitable base connecting means **154** may be employed. As non-limiting examples, the base **108** may be connected to the main body **106** using a plurality of threading connectors, thumb screws, friction fit, snap fit, or any other suitable means, as determined by one of skill in the art. The base **108** may be fabricated using any durable material or combination of materials such as metal, plastic, wood, and composite, as non-limiting examples. A skilled artisan may select any desirable shape and size for the base **108**.

(26) According to one more particular embodiment, the base **108** may include a plurality of legs **156** extending outwardly from the first end **116** of the main body **106** of the purging device **100** and configured to stabilize the purging device **100** with respect to the ground surface **124**. Each leg **156**

may be independently extendable, collapsible, and otherwise adjustable in both length and position relative to the ground surface **124** and the main body **106** of the purging device **100**. In one more particular embodiment, each leg **156** may be adjustable with respect to the angle at which the leg **156** extends outwardly from the main body **106** of the purging device **100**. According to certain more particular embodiments, each leg **156** may have a predetermined set of angled positions relative to the main body **106** of the purging device **100**.

(27) Advantageously, the adjustable legs **156** and easy-to-connect base **108** may allow the purging device **100** to be assembled and stabilized efficiently on uneven, rough ground surface **124**. Additionally, the adjustable legs **156** may be easily collapsed and compactly stored adjacent the main body **106** of the purging device **100**, as needed. The added stability added by the adjustable base **108** militates against hazardous conditions that may result from the purging device **100** falling over.

(28) With reference to FIG. **3**, the testing port **110** may include a first end **158** that is integrally formed with or permanently or removably connected to the sidewall **126** of the main body **106**. The testing port **110** may be in fluid communication with the main body **106** of the purging device **100**. The testing port **110** may have any desired shape, length, diameter, and thickness, as determined by one of skill in the art. Any material or combination of materials such as metal, plastic, composite, and wood, as non-limiting examples, may be used to form the testing port **110**. In certain embodiments, the testing port **110** may be fabricated using a material that is capable of withstanding line pressure up to or exceeding 125 PSI. Additionally, according to certain embodiments, a size, position, orientation, and configuration of the testing port **110** relative to the main body **106** of the purging device **100** may be configured to militate against the gas creating a Venturi effect.

(29) A testing valve **160** may be integrally formed in or connected to the testing port **110** device and configured to be selectively engaged by the technician to allow the gas **104** to flow out of a second end **162** of the testing port **110**. More specifically, the technician may selectively allow gas **104** to flow out of the second end **162** of the testing port **110** when the testing valve **160** is in an open position and prevent gas **104** from flowing out of the second end **162** of the testing port **110** when the testing valve **160** is in a closed position. A handle **164** can be used to alternate the testing valve **160** between the open position and the closed position. Any suitable testing valve **160** capable of permitting and preventing gas **104** flow from the second end **162** of the testing port **110** may be used, such as a ball valve, gate valve, globe valve, or butterfly valve, as non-limiting examples.

(30) In a more particular embodiment, the testing port **110** may be a 90-degree shoulder fitting including the testing valve **160**, as shown in FIG. **3**. In certain embodiments, the 90-degree shoulder fitting may be connected to the main body **106** using a three-eighths inch nipple welded to the main body **106**, as one non-limiting example. In another more particular embodiment, the testing port **110** may be positioned between the inlet **114** of the main body **106** and the outlet **118** of the main body **106**. The testing port **110** may militate against gas **104** creating a Venturi effect. It should be appreciated that one skilled in the art may employ different methods and structures for including a testing port **110** that is in fluid communication with the main body **106** of the purging device **100**.

(31) Advantageously, the testing port **110** allows the technician to measure the level of gas **104** within the gas system **102** while the gas **104** is flowing from the gas system **102** through the purging device **100**. Additionally, in certain embodiments, the testing port **110** is configured to militate against gas **104** creating a Venturi effect.

(32) The control valve **112** may be permanently or removably positioned within or along the main body **106** of the purging device **100** and configured to allow the technician to selectively prevent gas **104** from flowing out of the outlet **118** of the main body **106** using a handle **165** or other means extending outwardly from the main body **106** of the purging device **100**. Any suitable valve type

may be employed such as a ball valve, gate valve, globe valve, or butterfly valve, as non-limiting examples. In certain embodiments, the control valve **112** may be positioned within or along the main body **106** at a location that does not require the technician to bend over to access the control valve **112**. In one more particular embodiment, the control valve **112** may be positioned between the inlet **114** of the purging device **100** and the testing port **110**.
(33) It should be appreciated that a person skilled in the art may select any suitable control valve **112** and may adjust the position of the control valve **112** as required, within the scope of this disclosure. It should be further appreciated that a skilled artisan may employ any suitable structural components and methods for controlling the flow of gas **104** through the main body **106** of the purging device **100**, within the scope of this disclosure. Advantageously, the control valve **112** allows the technician to selectively allow gas **104** to be expelled from the purging device **100** or to prevent gas **104** from being expelled from the purging device **100**.
(34) In another embodiment of the present disclosure, a gas purging kit **166** is used to purge unwanted gas **104** from the gas system **102** or other gas source, as shown in FIG. **4**. The gas purging kit **166** includes the purging device **100** according to various embodiments of the present disclosure, at least one flexible hose **128**, at least one extender **140**, one or more safety cones **168**, and a carrying case **170**. In certain embodiments, the gas purging kit **166** may include tie-down straps or other securing means (not shown) and hook and loop, elastic, or other organizing connectors **172** for organizing cords, straps, and hoses. Additionally, safety signage, additional testing equipment, replacement parts, and various fittings and connectors (not shown) may be included, as determined by one of skill in the art.
(35) Advantageously, the gas purging kit **166** may be used to purge gas **104** or air from the gas system **102**, while militating against associated hazards. For example, the safety cones **168** may be placed around the purging device **100** and adjacent the flexible hose **128** to visually point out a possible tripping hazard associated with the purging device **100**. Additionally, the flexible hose **128** allows the technician to maintain a purge point at a safe operating distance. Likewise, the technician may adjust the overall height of the purging device **100** in order to meet safety standards and follow company specific procedures using one or more extenders **140**, as needed. In certain more particular embodiments, the gas purging kit **166** may include flexible hoses **128** of variable length as well as extenders **140** of variable height thereby allowing the technician to select and appropriate distance and height from the gas system **102**.
(36) The light weight, collapsible components of the gas purging kit **166** including the main body **106**, the base **108**, the flexible hose **128**, and the extender **140** are easy to store and transport using the convenient carrying case **170**. Advantageously, each of the base **108**, the flexible hose **128**, and the extender **140** may be collapsed and placed in a compact position alongside the main body **106** inside the carrying case **170**. The lightweight, thin materials such as aluminum and schedule 40 piping, as non-limiting materials, allow the technician to transport the gas purging kit **166** with ease. Likewise, the easy-to-use purge connecting means **132** and gas connecting means **136** allow for efficient set up and tear down of the gas purging kit **166**.
(37) In yet another embodiment of the present disclosure, with reference to FIG. **5**, a method **400** of using the gas purging kit **166**, is shown. The method **400** may include a first step **402** of providing the gas purging kit **166**. The method **400** may further include a second step **404** of identifying the gas system **102** for purging. A third step **406** may include assembling the purging device **100**. As one non-limiting example, assembling the purging device **100** may include connecting the main body **106** to the base **108**, adjusting the legs **156** of the base **108**, selecting one or more of the flexible hose **128** and the extender **140** for use in combination with the purging device **100**, securing tie-down straps to the securing elements **148** and to the ground surface **124**, and securing the rod and cable **150** to the ground surface **124**, as non-limiting examples. An additional step **408** may include connecting the inlet **114** of the purging device **100** to the gas system **102** using the flexible hose **128** such that the purging device **100** and the gas system **102** are in fluid

communication with one another. A final step **410** may include purging the gas **104** from the gas system **102**.
(38) It should be appreciated that additional steps may be included, as determined by one of skill in the art. As non-limiting examples, various steps relating to the setup of the purging device **100**, adjustment of various components such as the plurality of legs **156**, safety precautions, testing, and disassembling of the gas purging kit **166** may be included. The method **400** for using the gas purging kit **166** may also include repeating or omitting various steps, as required. Advantageously, the purging device **100**, the gas purging kit **166**, and the method **400** of using the gas purging kit **166** may be used in a multitude of settings and may be customized, as determined by a skilled artisan.
(39) Example embodiments are provided so that this disclosure will be through and fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms, and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail. Equivalent changes, modifications and variations of some embodiments, materials, components and methods may be made within the scope of the present technology, with substantially similar results.

Claims

1. A purging device, comprising: a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body, the inlet including a conduit extending outwardly at a 90-degree angle from a sidewall of the main body; a base including a plurality of legs, each leg of the plurality of legs pivotally connected to a lowermost portion of the first end of the main body wherein the base is disposed below the inlet; a testing port disposed along the main body of the purging device, the testing port including a shoulder fitting configured to position the testing port substantially parallel to the main body; and a control valve disposed between the inlet of the main body and the outlet of the main body, the testing port disposed between the second end of the main body and the control valve.
2. The purging device of claim 1, wherein the main body is fabricated from schedule 40 piping.
3. The purging device of claim 1, further comprising a flexible hose adapted to connect to the inlet of the main body and provide fluid communication between the purging device and a gas system.
4. The purging device of claim 3, wherein the flexible hose is at least 12 feet in length.
5. The purging device of claim 1, wherein an extender is adapted to connect to the outlet of the main body.
6. The purging device of claim 5, wherein the extender is fabricated from aluminum.
7. The purging device of claim 5, wherein a height of the main body combined with a height of the extender is between seven feet and ten feet.
8. The purging device of claim 1, wherein the base has a plurality of securing elements extending outwardly from a sidewall of the main body, the securing elements configured to secure the purging device to a ground surface.
9. The purging device of claim 1, wherein the purging device includes a grounding rod and cable connected to the main body.
10. The purging device of claim 1, wherein the base is removably connected to the first end of main body.
11. The purging device of claim 1, wherein a length of each leg of the plurality of legs is independently adjustable.

12. The purging device of claim 1, wherein each leg of the plurality of legs extends outwardly from the main body of the purging device, and wherein an angle of each leg relative to the main body of the purging device is adjustable.
13. The purging device of claim 1, wherein the testing port extends outwardly from a sidewall of the main body, the testing port in fluid communication with the main body.
14. The purging device of claim 1, wherein each one of the legs each leg of the plurality of legs is disposed orthogonally to the main body.
15. The purging device of claim 1, wherein the lowermost portion of the first end of the main body further includes a projection, each leg of the plurality of legs pivotally connected to the projection.
16. The purging device of claim 1, wherein the shoulder fitting of the testing port is a 90-degree shoulder fitting.
17. A gas purging kit, comprising: a purging device including a main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body, the inlet including a conduit extending outwardly at a 90-degree angle from a sidewall of the main body, a base including a plurality of legs, each leg of the plurality of legs pivotally connected to a lowermost portion of the first end of the main body wherein the base is disposed below the inlet, a testing port disposed along the main body of the purging device, the testing port including a shoulder fitting configured to position the testing port substantially parallel to the main body, and a control valve disposed between the inlet of the main body and the outlet of the main body, the testing port disposed between the second end of the main body and the control valve; a flexible hose adapted to connect to the inlet of the main body; an extender adapted to connect to the outlet of the main body; at least one safety cone; and a carrying case.
18. The gas purging kit of claim 17, wherein each leg of the plurality of legs is independently adjustable, and an angle of each leg of the plurality of legs relative to the main body of the purging device is adjustable.
19. The gas purging kit of claim 17, wherein the testing port includes a 90-degree shoulder fitting and the testing port is adapted to militate against a gas from a gas system creating a Venturi effect.
20. A method of using a gas purging kit, comprising: providing a gas purging kit including a purging device having main body having an inlet disposed at a first end of the main body and an outlet disposed at a second end of the main body, the inlet including a conduit extending outwardly at a 90-degree angle from a sidewall of the main body, a base including a plurality of legs, each leg of the plurality of legs pivotally connected to a lowermost portion of the first end of the main body wherein the base is disposed below the inlet, a testing port disposed along the main body of the purging device, the testing port including a shoulder fitting configured to position the testing port substantially parallel to the main body, and a control valve disposed between the inlet of the main body and the outlet of the main body, the testing port disposed between the second end of the main body and the control valve, a flexible hose adapted to connect to the inlet of the main body, an extender adapted to connect to the outlet of the main body, at least one safety cone, and a carrying case; identifying a gas system for purging; assembling the purging device; connecting the inlet of the purging device to the gas system; and purging a gas from the gas system.